

Analysis of OMICS data

Lecture #9 - ChIP-seq analysis

17apr-2026



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the Central Dogma of Molecular Biology

genes

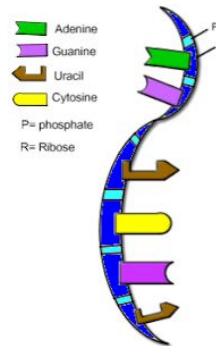


genomics

**what could the
cell do?**



mRNA

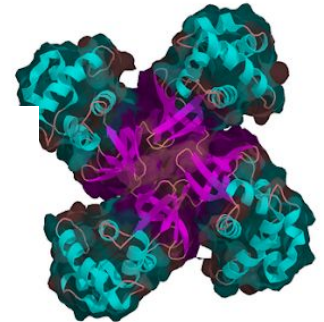


transcriptomics

**what is the cell
going to do?**



proteins



proteomics

**what is the cell
actually doing?**

~~the Central Dogma of Molecular Biology~~

the five omics

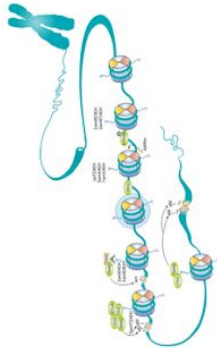
genes



genomics

what could the cell do?

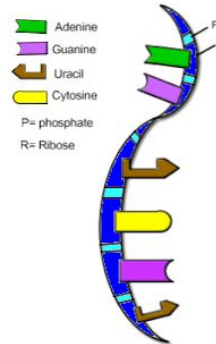
chromatin



epigenomics

what intends the cell to do?

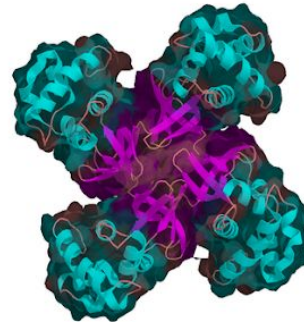
mRNA



transcriptomics

what is the cell going to do?

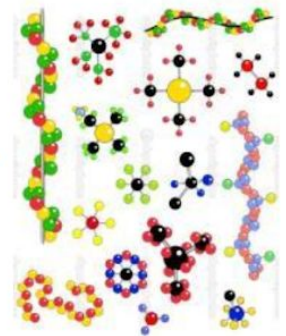
proteins



proteomics

what is the cell actually doing?

metabolites



metabolomics

what did the cell do?

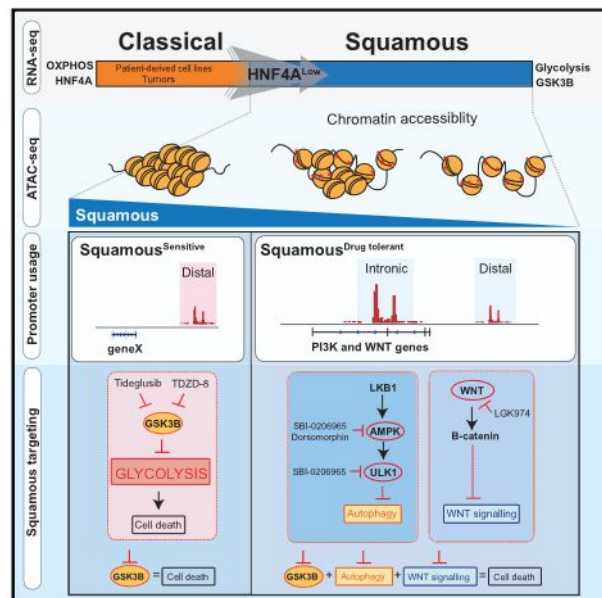
Some epigenomic wins

Cell Reports

Article

HNF4A and GATA6 Loss Reveals Therapeutically Actionable Subtypes in Pancreatic Cancer

Graphical Abstract



Authors

Holly Brunton, Giuseppina Caligiuri, Richard Cunningham, ..., David K. Chang, Peter J. Bailey, Andrew V. Biankin

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In Brief

Brunton et al. demonstrate that differential chromatin accessibility can predict responsiveness and tolerance to GSK3 β inhibitors in the squamous subtype of PDAC. This study provides an important proof of concept that chromatin accessibility can be used to identify additional PDAC subgroups with potential therapeutic utility.

#1 ATAC-seq revealed clinically relevant subtypes of pancreatic cancer and drug-response states that RNA alone did not separate.

<https://www.sciencedirect.com/science/article/pii/S2211124720305787> -> «HNF4A and GATA6 Loss Reveals Therapeutically Actionable Subtypes in Pancreatic Cancer» (2020)

Some epigenomic wins

#2 plasma
epigenomic
cell-free DNA
non-invasively
detected lung
adenocarcinoma
transforming into
small-cell lung
cancer.

CLINICAL CANCER RESEARCH | PRECISION MEDICINE AND IMAGING

Detecting Small Cell Transformation in Patients with Advanced *EGFR* Mutant Lung Adenocarcinoma through Epigenomic cfDNA Profiling

Talal El Zarif^{1,2,3}, Catherine B. Meador^{4,5}, Xintao Qiu³, Ji-Heui Seo^{1,2}, Matthew P. Davidsohn^{1,2}, Hunter Savignano^{1,2}, Gitanjali Lakshminarayanan^{1,2}, Heather M. McClure^{1,2}, John Canniff^{1,2}, Brad Fortunato^{1,2}, Rong Li², Mandeep K. Banwait^{4,5}, Karl Semaan^{1,2,6}, Marc Eid^{1,2}, Henry Long², Yin P. Hung⁷, Navin R. Mahadevan^{1,8}, David A. Barbie¹, Matthew G. Oser¹, Zofia Piotrowska^{4,5}, Toni K. Choueiri¹, Sylvan C. Baca^{1,2,6}, Aaron N. Hata^{4,5}, Matthew L. Freedman^{1,2,6}, and Jacob E. Berchuck¹



ABSTRACT

Purpose: Histologic transformation to small cell lung cancer (SCLC) is a mechanism of treatment resistance in patients with advanced oncogene-driven lung adenocarcinoma (LUAD) that currently requires histologic review for diagnosis. Herein, we sought to develop an epigenomic cell-free DNA (cfDNA)-based approach to noninvasively detect small cell transformation in patients with *EGFR* mutant (*EGFR*m) LUAD.

Experimental Design: To characterize the epigenomic landscape of transformed (t)SCLC relative to LUAD and *de novo* SCLC, we performed chromatin immunoprecipitation sequencing (ChIP-seq) to profile the histone modifications H3K27ac, H3K4me3, and H3K27me3; methylated DNA immunoprecipitation sequencing (MeDIP-seq); assay for

1 mL aliquots of plasma from patients with *EGFR*m LUAD with or without tSCLC.

Results: Analysis of 126 epigenomic libraries from the lung cancer PDXs revealed widespread epigenomic reprogramming between LUAD and tSCLC, with a large number of differential H3K27ac ($n = 24,424$), DNA methylation ($n = 3,298$), and chromatin accessibility ($n = 16,352$) sites between the two histologies. Tumor-informed analysis of each of these three epigenomic features in cfDNA resulted in accurate noninvasive discrimination between patients with *EGFR*m LUAD versus tSCLC [area under the receiver operating characteristic curve (AUROC) = 0.82–0.87]. A multianalyte cfDNA-based classifier integrating these three epigenomic features discriminated

<https://pubmed.ncbi.nlm.nih.gov/38912901/> -> «Detecting Small Cell Transformation in Patients with Advanced *EGFR* Mutant Lung Adenocarcinoma through Epigenomic cfDNA Profiling» (2024)

Some epigenomic wins

Mod Pathol 39 (2026) 100945

MODERN PATHOLOGY



Journal homepage: <https://modernpathology.org/>

Research Article

Diagnostic Utility and Clinicopathologic Associations of Histone H3 Lysine 27 Trimethylation (H3K27me3) Immunohistochemistry for Merkel Cell Carcinoma

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^a Department of Pathology, University of Michigan Medical School, Ann Arbor, Michigan; ^b Department of Dermatology, University of Michigan Medical School, Ann Arbor, Michigan; ^c Rogel Cancer Center, University of Michigan Medical School, Ann Arbor, Michigan; ^d Michigan Center for Translational Pathology, University of Michigan Medical School, Ann Arbor, Michigan

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ABSTRACT

Merkel cell carcinoma (MCC) is an aggressive cutaneous tumor that must be distinguished from other cutaneous tumors and metastatic small cell carcinoma (SmCC). Additional diagnostic markers are limited for MCC with immunophenotypic aberrancy. MCC can display immunohistochemical loss of the epigenetic marker histone H3 lysine 27 trimethylation (H3K27me3), but to our knowledge, the diagnostic utility of this observation has not been evaluated. In this study, we investigate H3K27me3 labeling in MCC (n = 195), cutaneous epithelial tumors (n = 48), non-cutaneous SmCCs (n = 56), and olfactory neuroblastoma (n = 11), comparing with diagnostic markers, cytokeratin-20, neurofilament, SATB2, and POU4F3. H3K27me3 patterns in MCC included global loss, variable/mosaic labeling, and diffuse labeling. Global loss significantly associated with

#3 H3K27me3 loss served as a practical diagnostic marker for an aggressive type of skin cancer, showing the histone mark itself can be clinically useful.

<https://pubmed.ncbi.nlm.nih.gov/41352627/> -> «Diagnostic Utility and Clinicopathologic Associations of Histone H3 Lysine 27 Trimethylation (H3K27me3) Immunohistochemistry for Merkel Cell Carcinoma» (2025/2026)

Some epigenomic wins

#4 comparative 3D epigenomics showed chromatin loops are an ancestral feature of animal regulatory genomes, arising before bilaterian diversification.

Article

Chromatin loops are an ancestral hallmark of the animal regulatory genome

<https://doi.org/10.1038/s41586-025-08960-w>

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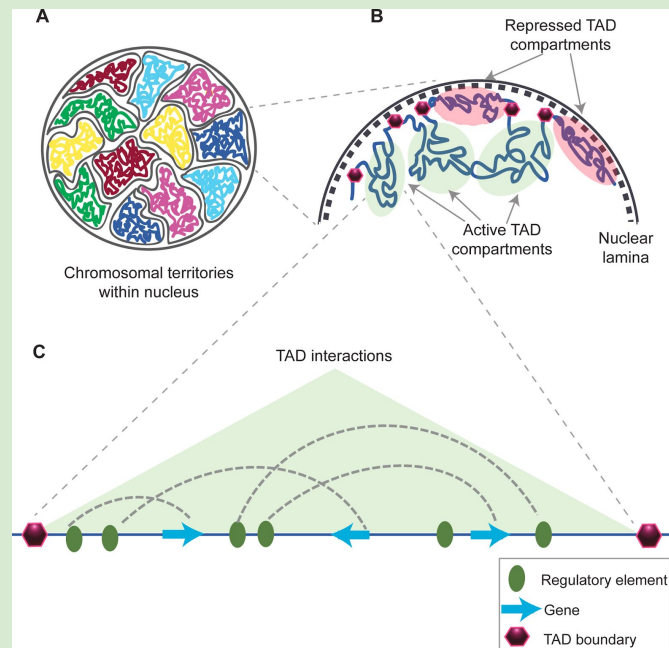
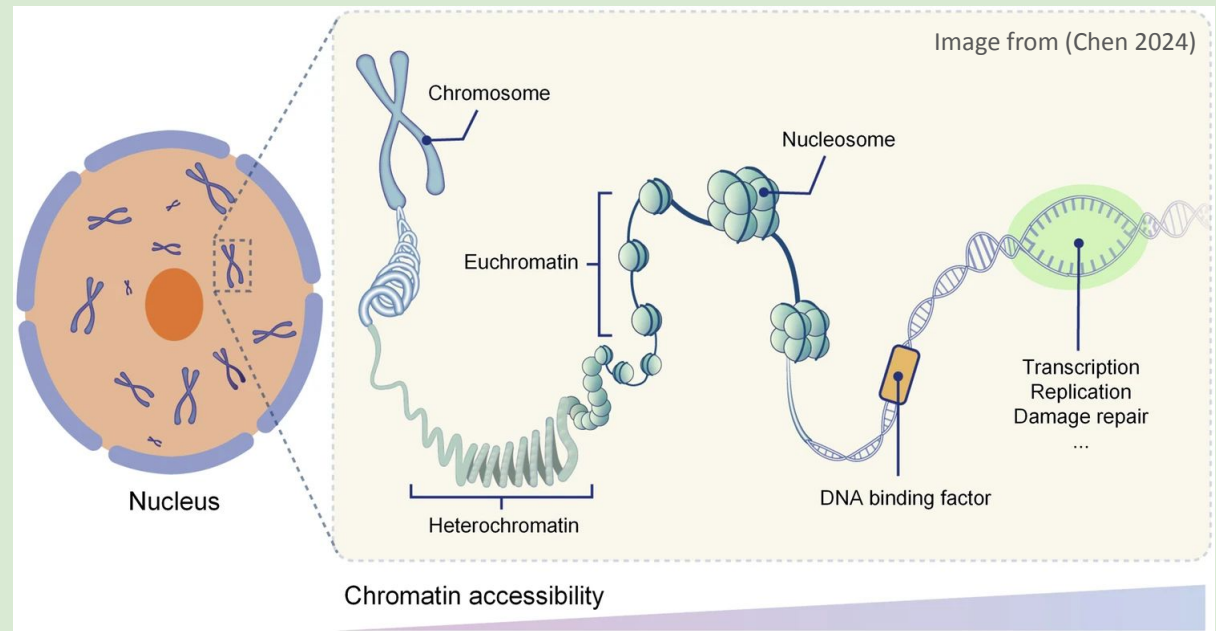
 Check for updates

Iana V. Kim^{1,2,3}, Cristina Navarrete^{1,3,10}, Xavier Grau-Bové^{1,10}, Marta Iglesias¹, Anamaria Elek¹, Grygoriy Zolotarov¹, Nikolai S. Bykov², Sean A. Montgomery¹, Ewa Ksiezopolska¹, Didac Cañas-Armenteros¹, Joan J. Soto-Angel⁴, Sally P. Leys⁵, Pawel Burkhardt⁴, Hiroshi Suga⁶, Alex de Mendoza⁷, Marc A. Martí-Renom^{1,2,3,8} & Arnau Sebé-Pedrós^{1,3,8,9}✉

In bilaterian animals, gene regulation is shaped by a combination of linear and spatial regulatory information. Regulatory elements along the genome are integrated into gene regulatory landscapes through chromatin compartmentalization^{1,2}, insulation of neighbouring genomic regions^{3,4} and chromatin looping that brings together distal *cis*-regulatory sequences⁵. However, the evolution of these regulatory features is unknown because the three-dimensional genome architecture of most animal lineages remains unexplored^{6,7}. To trace the evolutionary origins of animal genome regulation, here we characterized the physical organization of the genome in non-bilaterian animals (sponges, ctenophores, placozoans and cnidarians)^{8,9} and their closest unicellular relatives (ichthyosporeans, filastereans and choanoflagellates)¹⁰ by combining high-resolution chromosome conformation capture^{11,12} with epigenomic marks and gene expression data. Our comparative analysis showed that chromatin looping is a conserved feature of genome architecture in ctenophores, placozoans and cnidarians. These sequence-determined distal contacts involve both promoter–enhancer and promoter–promoter interactions. By contrast, chromatin loops are absent in the unicellular relatives of animals. Our findings indicate that spatial

<https://www.nature.com/articles/s41586-025-08960-w> ->
«Chromatin loops are an ancestral hallmark of the animal regulatory genome» (2025)

DNA is not a
naked, flat
molecule, it
has 3D
structure,
histones,
transcription
factors and
domains



same DNA

BUT!!!

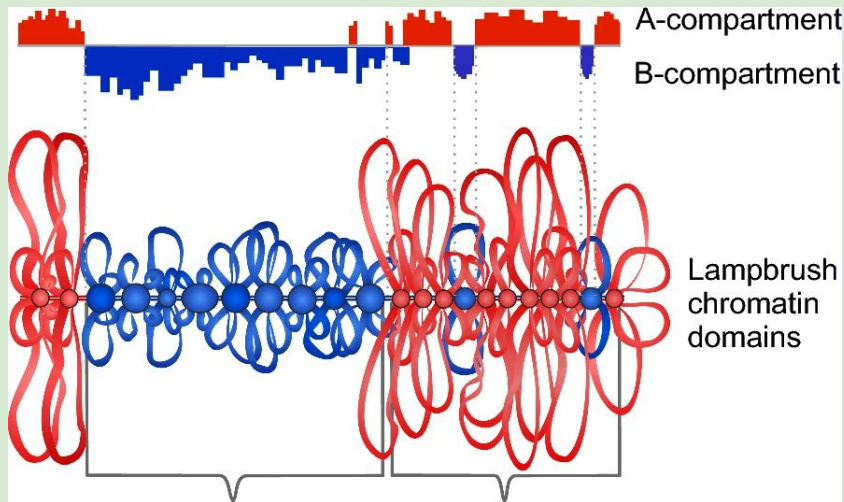
**100+ cell
types!**

Topologically Associated
Domains (TAD)

Image from (Matharu 2015)

A and B compartments large-scale active vs inactive chromatin neighbourhoods.

A compartments	B compartments
more open/active	more compact/repressed
gene-rich	lamina-associated
≈ euchromatin	≈ heterochromatin

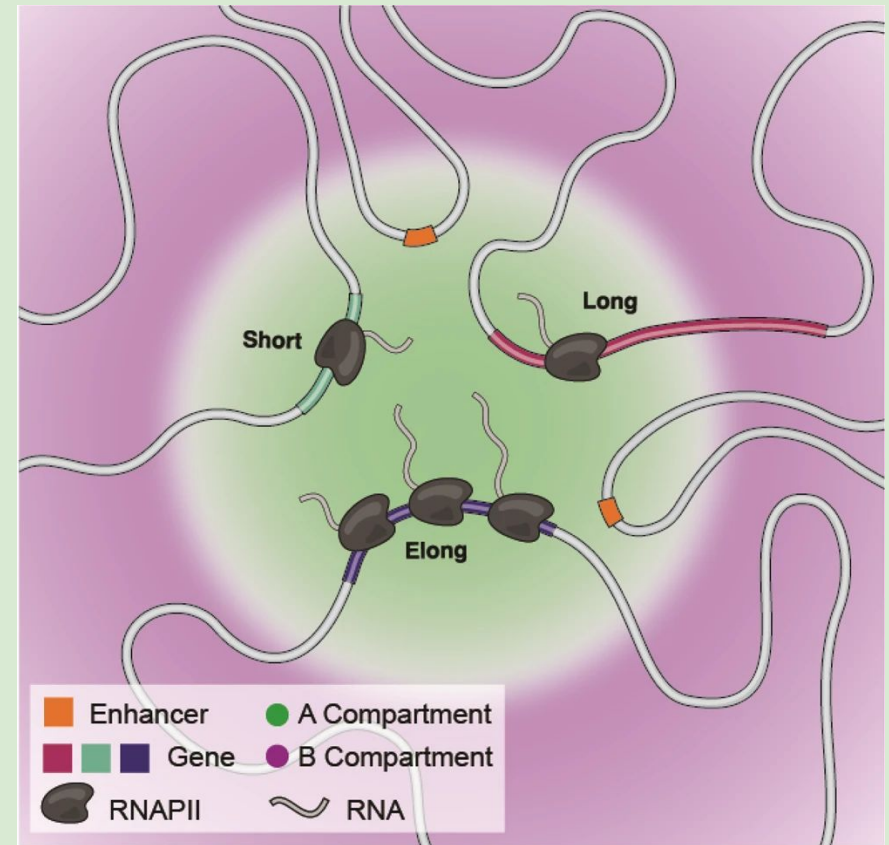


Cluster of large and dense chromomeres with short lateral loops

Chromatin:
- closed
- condensed
- large portion is transcriptionally silent

Cluster of small and loose chromomeres with long lateral loops

Chromatin:
- open
- decondensed
- transcriptionally active
- RNA-pol II enriched



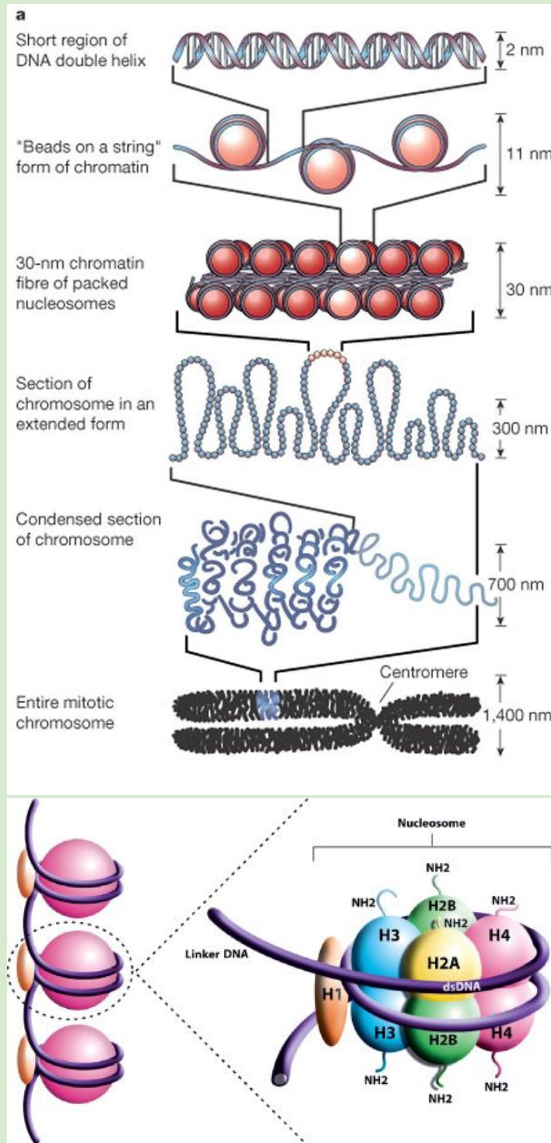
Source (Harris 2023)

Compartments are large-scale features (kilobase scale), larger than the usual scale for ChIP-seq

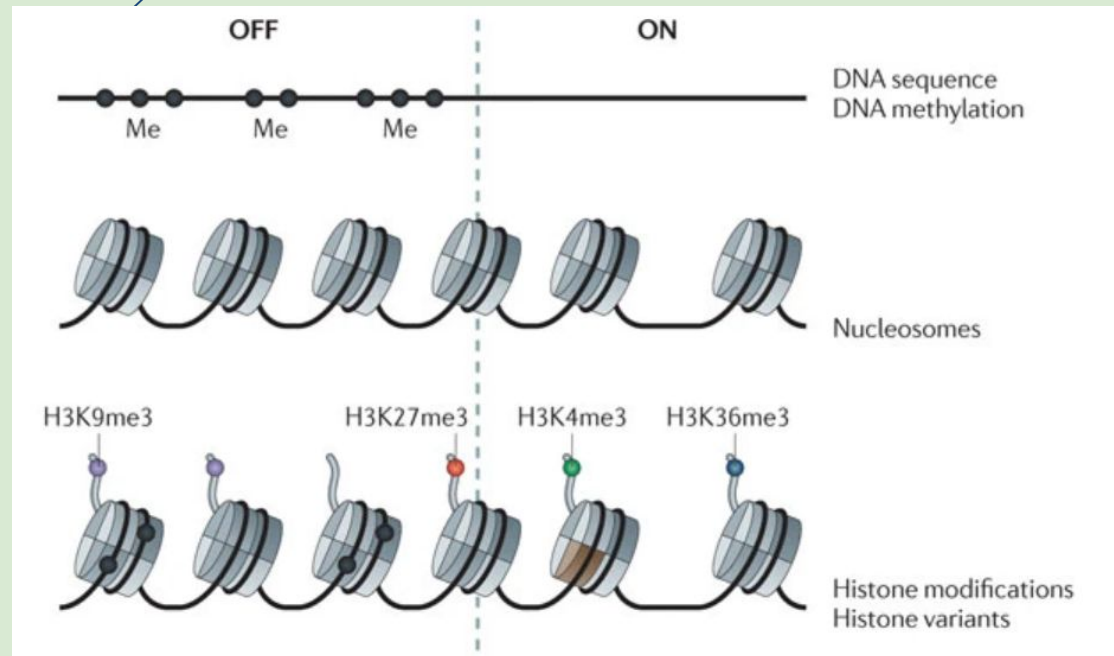
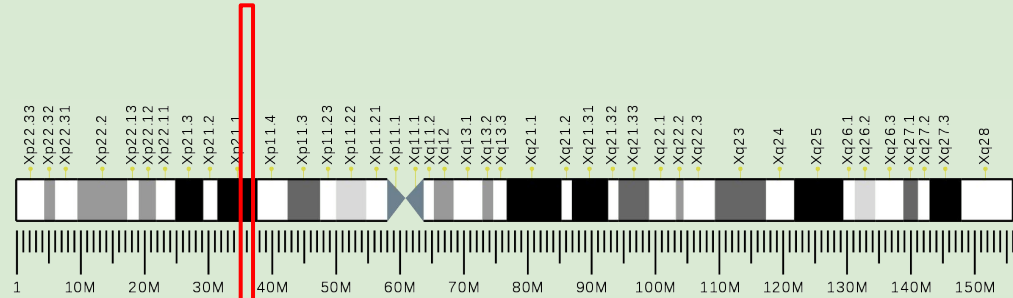
What kind of chromatin? > euchromatin/heterochromatin
Where in 3D neighbourhood? > A/B compartments

Source (Krasikova 2023)

Histone modifications



Source (Felsenfeld 2003)



Source (Villalta 2007)

Source (Zhou 2011)

post translational modifications on histones often mark different chromatin activities

H3K27ac > active enhancers/promoters

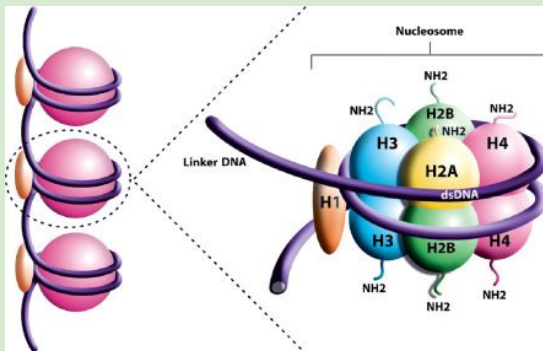
H3K4me1 > enhancer-like regions

H3K27me3 > Polycomb-repressed chromatin

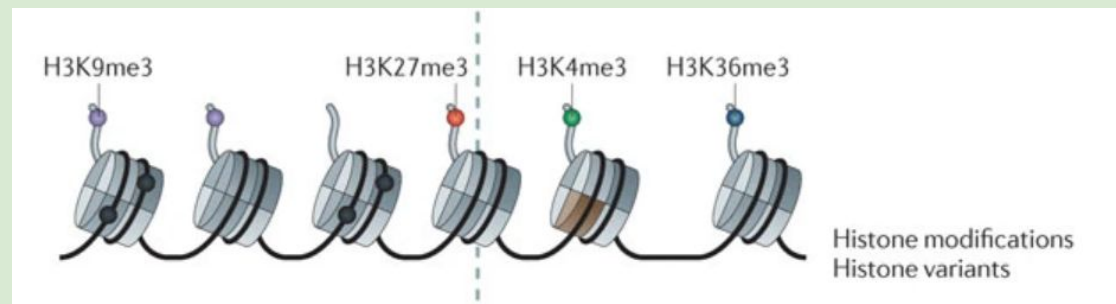
H3K4me3 > active promoters/TSSs

H3K9me3 > constitutive heterochromatin

H3K36me3 > transcribed gene bodies



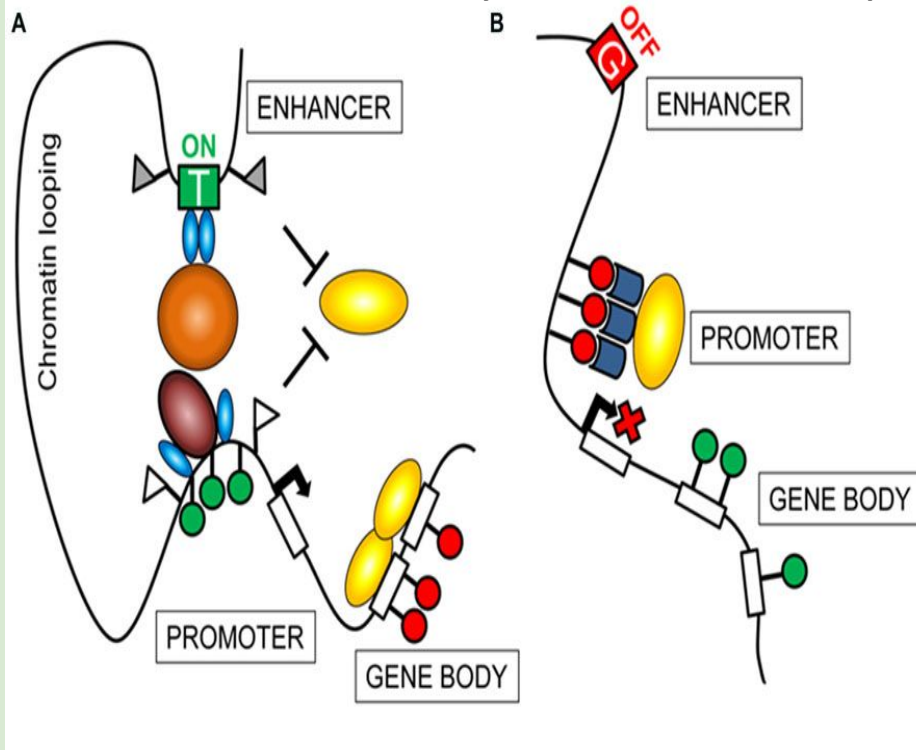
Source (Villalta 2007)



Source (Zhou 2011)

Two especially interesting in combination are **H3K27ac** (activity of enhancers/promoters) and **H3K4me1** (enhancer)

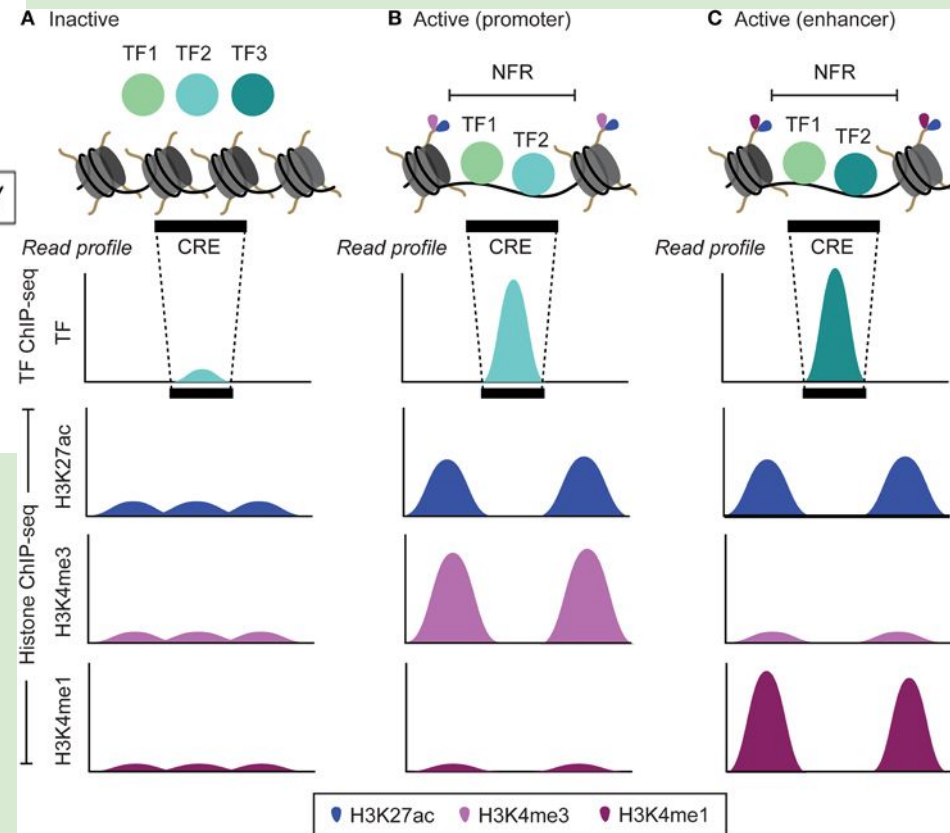
**We will consider these
in our practical!**



(Heyn 2014)

So...

- Active promoter: H3K4me3 + H3K27ac.
- Active enhancer: H3K4me1 + H3K27ac.



(Pundhir 2015)

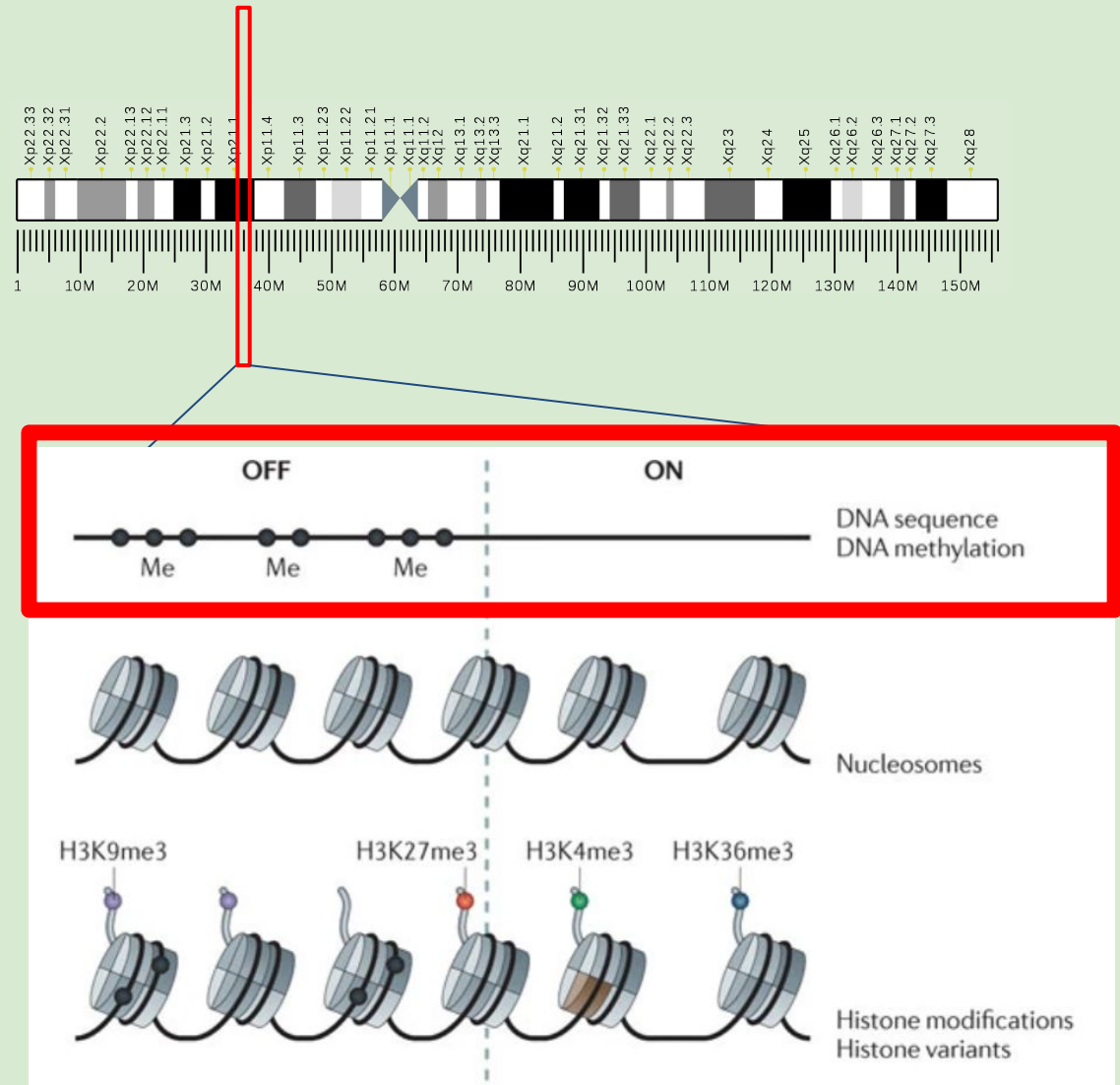
DNA methylation

Complementary
epigenomic layer

Often linked to
stable repression
and cell identity

Not analysed with
ChIP-seq

But many
computational models
are the same



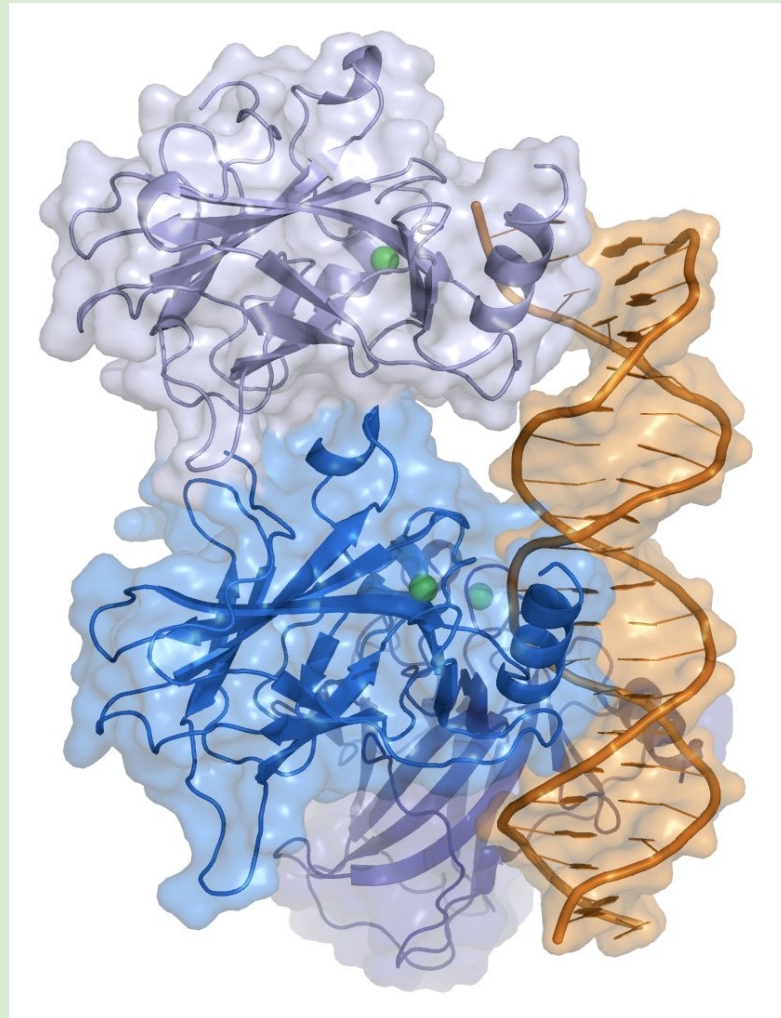
Transcription factors

They bind sequence motifs

Binding depends on chromatin accessibility and regulatory context

Their binding sites can be identified with ChIP-seq

Very important to unveil gene expression patterns



Polycomb repression and H3K27me3

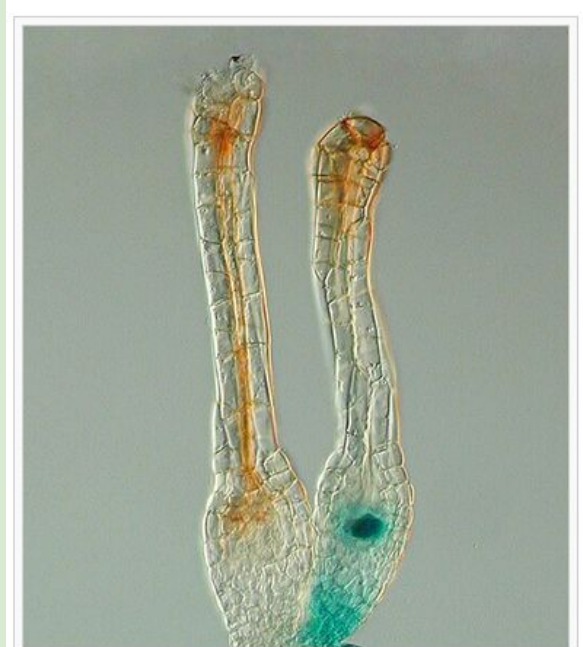
Not all silence is the same! Polycomb repression is a regulated chromatin state

Polycomb proteins (PcG proteins) are a family of protein complexes that specialises in the **epigenetic silencing** of genes.

Some act by modifying histones to H3K27me3 (PRC2), a label that other polycomb proteins (PRC1) use to pack DNA tightly.

They illustrate that silenced chromatin can also be highly regulated and biologically informative.

Polycomb silencing is often related to development, lineage choice and cell-identity.



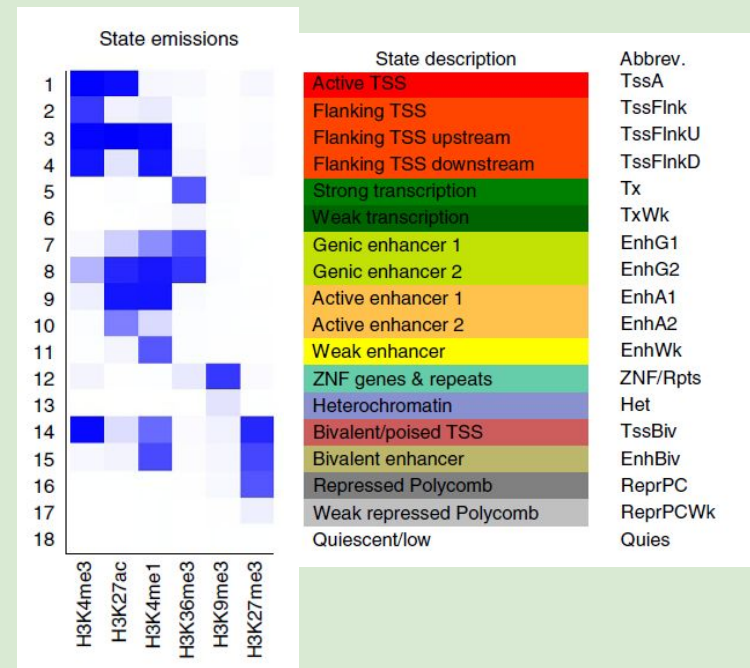
The Polycomb gene FIE is expressed (blue) in unfertilised egg cells of the moss *Physcomitrella patens* (right) and expression ceases after fertilisation in the developing diploid sporophyte (left). In situ GUS staining of two female sex organs (archegonia) of a transgenic plant expressing a translational fusion of FIE-uidA under control of the native FIE promoter

Chromatin states

emerge from combinations of histone marks

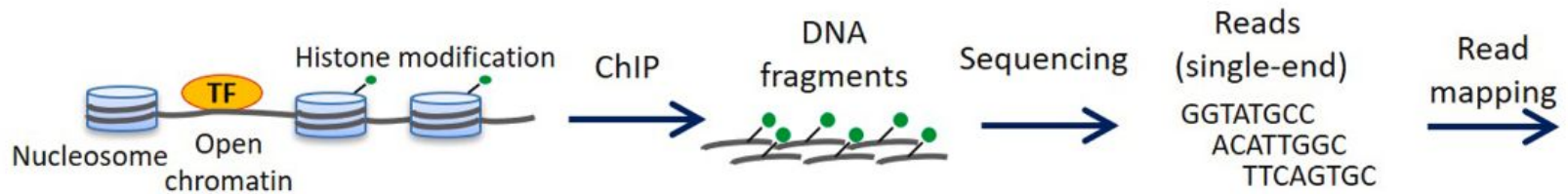
Presence/absence of different marks in the same genomic bin can encode different **chromatin states**

In the practical we will see this when using **ChromHMM**

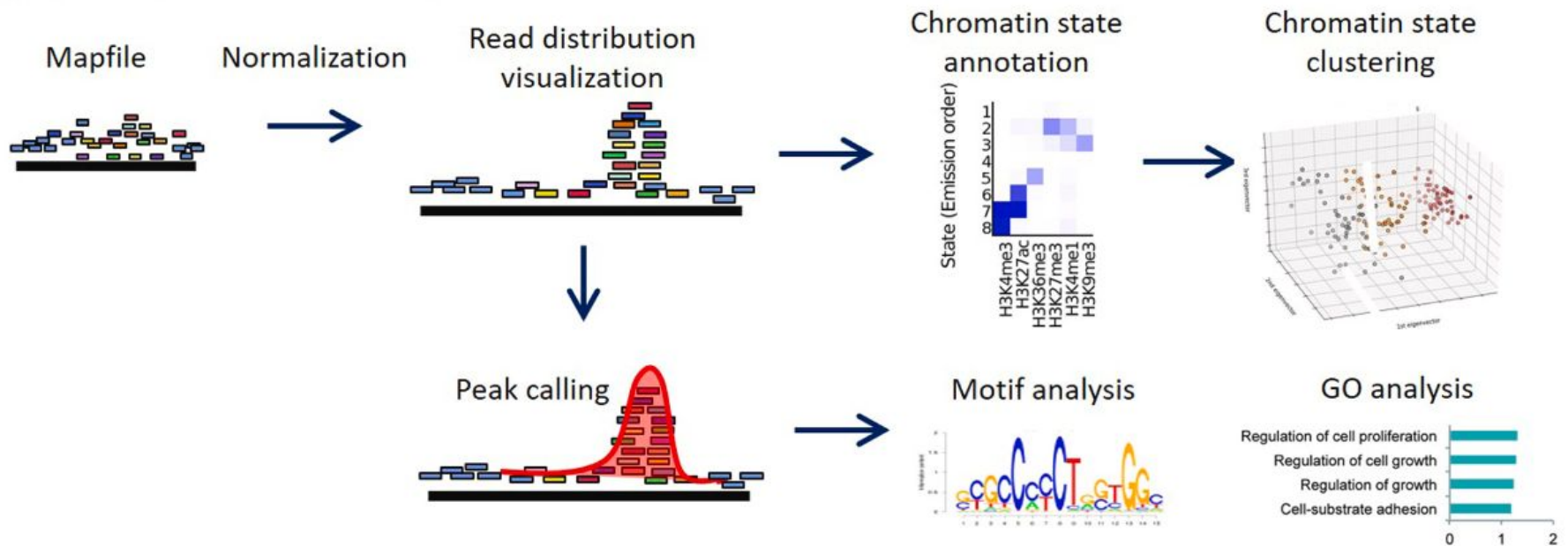


ChIP-seq workflow

(A) Sample preparation and sequencing



(B) Computational analysis

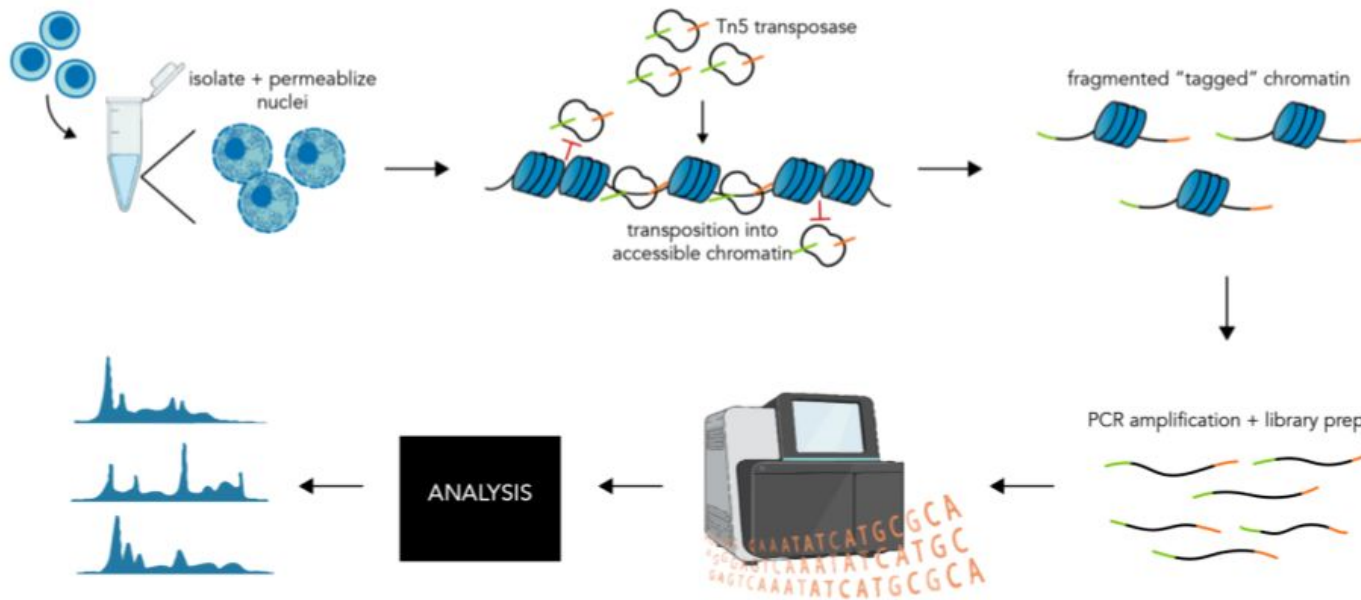


ATAC-seq

(Assay for Transposase-Accessible Chromatin using sequencing)

ATAC-seq maps open chromatin (accessible regions)

Overview of ATAC-seq



Source:
https://hutchdatascience.org/Choosing_Genomics_Tools/atac-seq-1.html

ChIP-seq is not the only assay

ChIP-seq measures proteins binding to DNA, but it is often used along with:

RNA-seq -> transcriptomics

WGS (whole genome sequencing)

DNA methylation

ATAC-seq -> measures accessibility

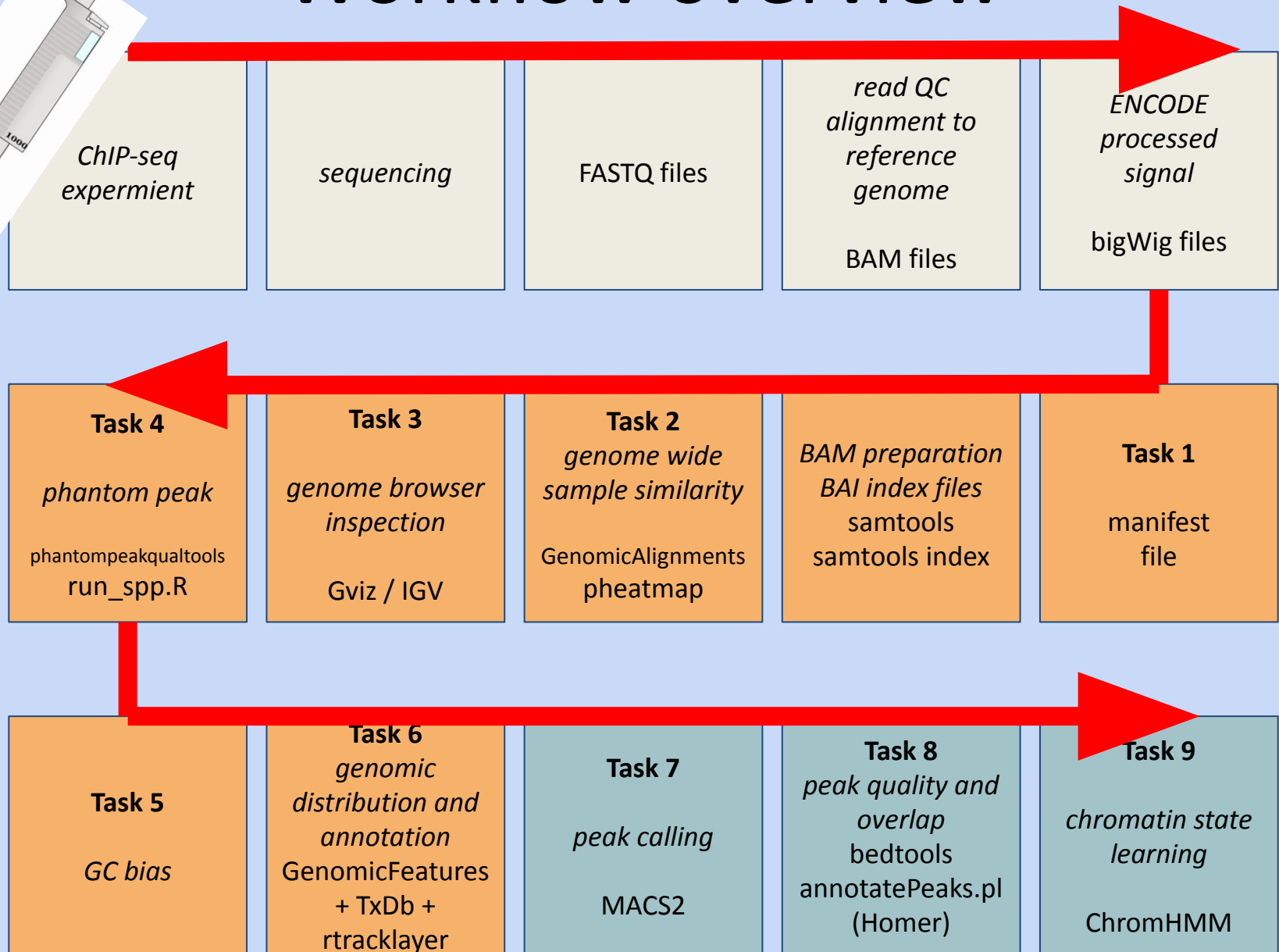
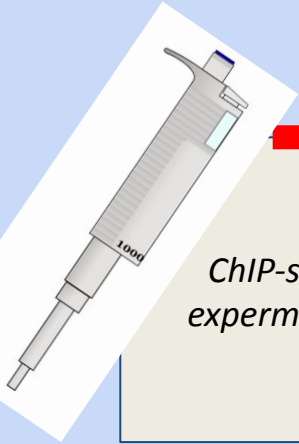
Hi-C -> to reveal 3D contacts in chromatin

There are also other assays similar to ChIP-seq such as:

CUT & RUN -> includes “scissors” (an enzyme) to cut the DNA next to the antibody

CUT & Tag -> includes enzymes that attach adapters (a small DNA sequence) that makes the piece easy to sequence

Workflow overview



What are exactly our samples and files?

We will use a dataset from a leukemia cell line: K562

Files:

H3K27ac

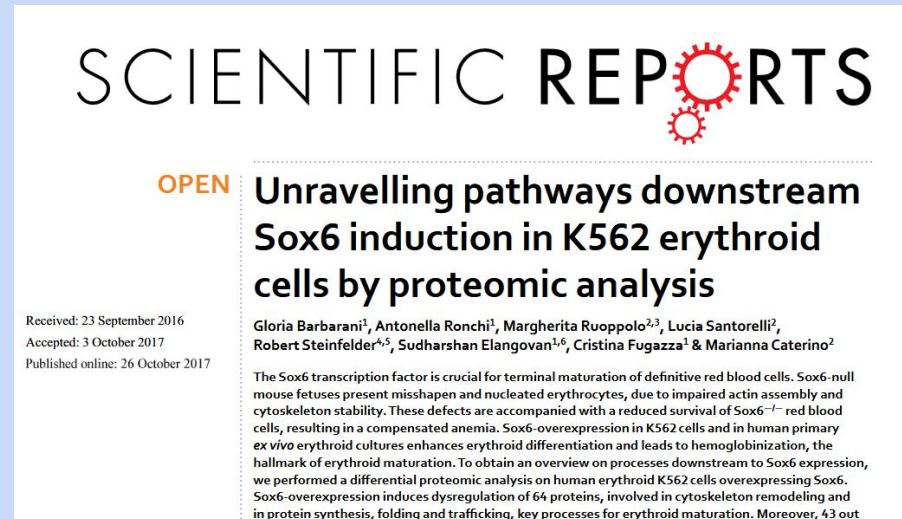
H3K4me1

shared Input.

Formats:

BAM = aligned reads for counting and peak calling

bigWig = processed signal track for browser inspection



Experiments / ChIP-seq / Homo sapiens / K562

Experiment summary for ENCSR000AKP

doi:10.17989/ENCSR000AKP

1 9

Summary	Attribution
Status: released	Lab: Bradley Bernstein, Broad
Assay: ChIP-seq (Histone ChIP-seq)	Award: U54HG004570 (Bradley Bernstein, Broad)
Target: H3K27ac	Project: ENCODE
Biosample summary: Homo sapiens K562	External resources: UCSC-ENCODE-hg19:wgEncodeEH000043 GEO:GSM733656
Biosample Type: cell line	References: doi:10.1038/s41467-020-14743-w PMID:32728046 PMCID:PMC7391744 doi:10.1038/s41592-020-0907-8 PMID:32737473 PMCID:PMC8073243 PMCID:PMC7343189 PMCID:PMC6724667
Replication type: isogenic	Aliases: bradley-bernstein:Project Element 91, bradley-bernstein:Project Element 92
Description: H3K27ac ChIP-seq on human K562	
Nucleic acid type: DNA	
Size range: 200-600	
Fragmentation: see document	

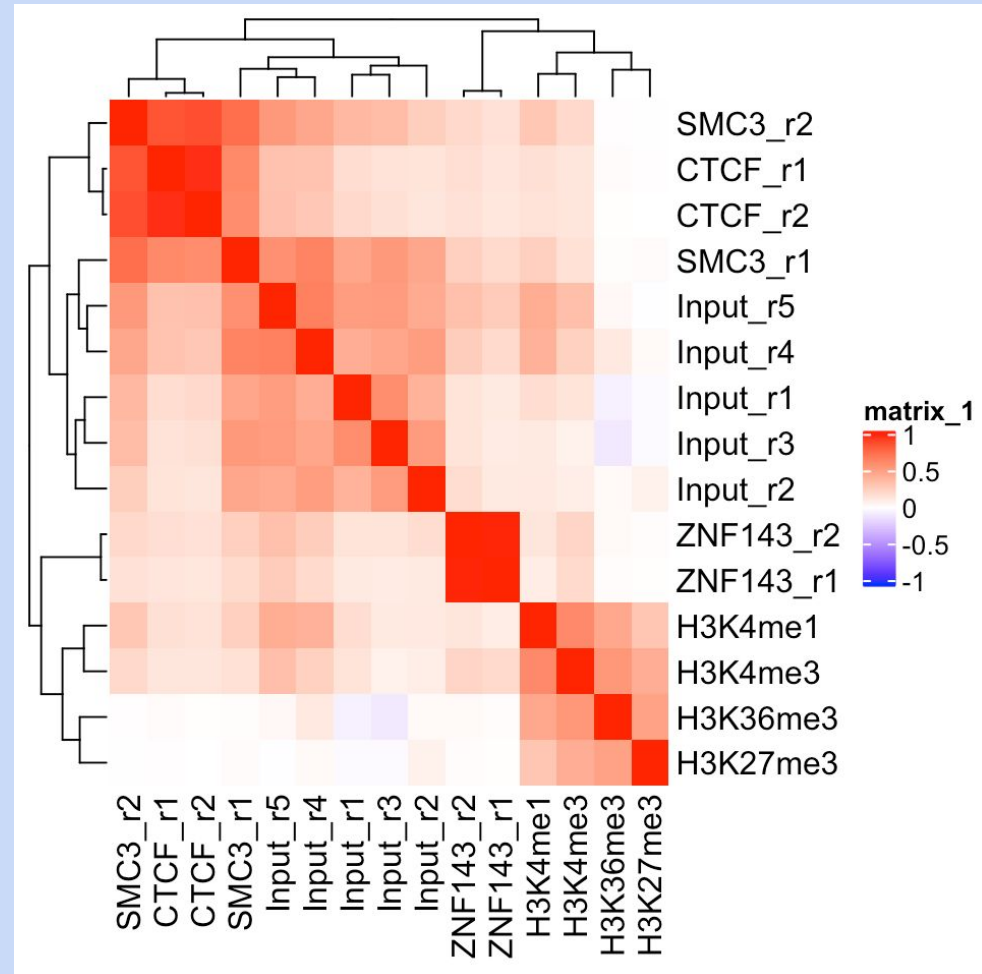
First QC: do the samples show a sensible genome-wide pattern?

Algorithm:

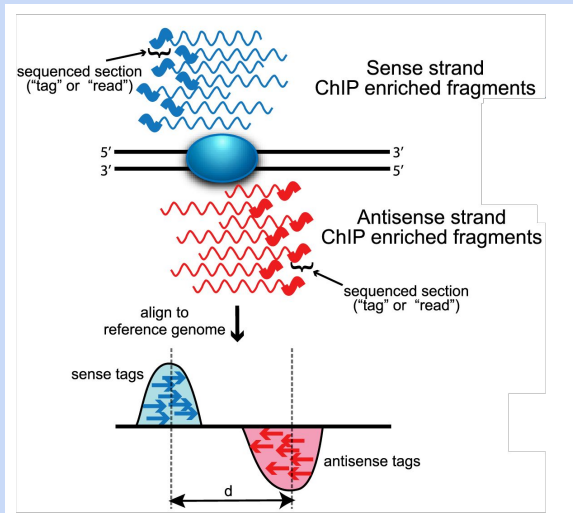
- count reads in 1-kb bins
- normalising
- create a correlation matrix
- clustering samples

Akalin uses many samples, and gets this correlation matrix.

We will have only two samples + shared input, but essentially is the same.



Strand cross-correlation and phantom peak



Source (Wilbanks and Facciotti 2010)

In ChIP-seq we don't get the exact protein-binding centre, we usually sequence just a bit of the 5' side from each DNA strand. By readjusting the position of each (the **strand cross-correlation**), we calculate the true ChIP enrichment signal (fragment-length peak). But sometimes there is a mismatch (which creates the phantom peak).

fragment-length peak: true ChIP enrichment signal

phantom peak: a read-length cross-correlation artefact

We calculate two parameters to control this:

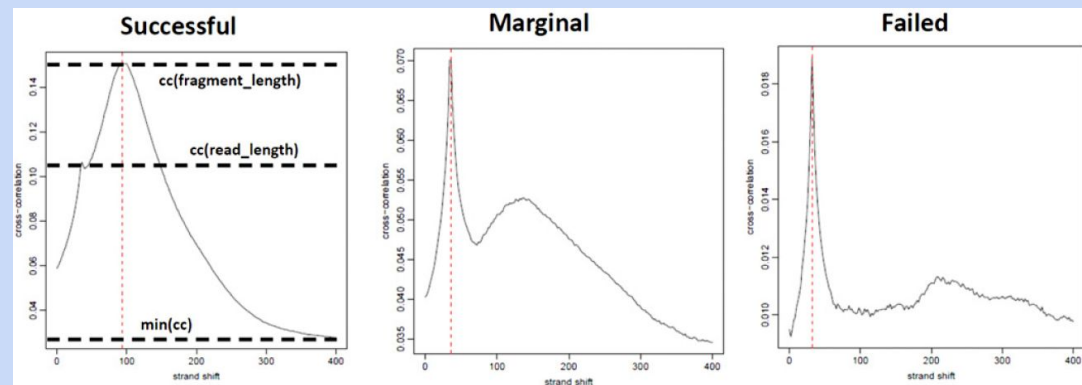
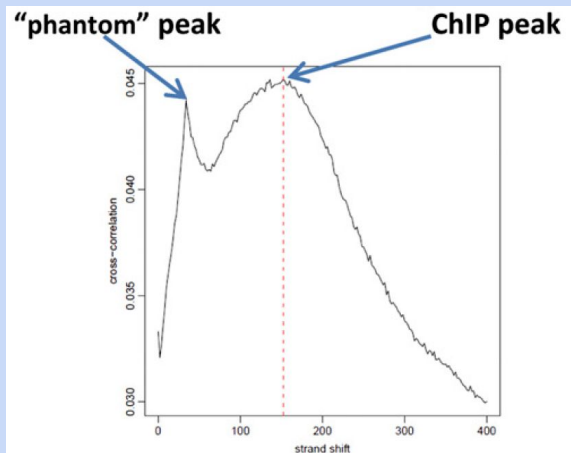
NSC (Normalised Strand Cross-correlation coefficient) = fragment peak vs background

RSC (Relative Strand Cross-correlation coefficient) = fragment peak vs phantom peak

Fragment-length peak > phantom peak (NSC > 1.05, RSC > 0.8)

Yes? > Good to go!

No? > Oops!



Source (Landt 2012)

GC bias

signal... or just noise
due too much/too little GC?

G≡C pairs have
three hydrogen
bonds
A=T pairs have
only two

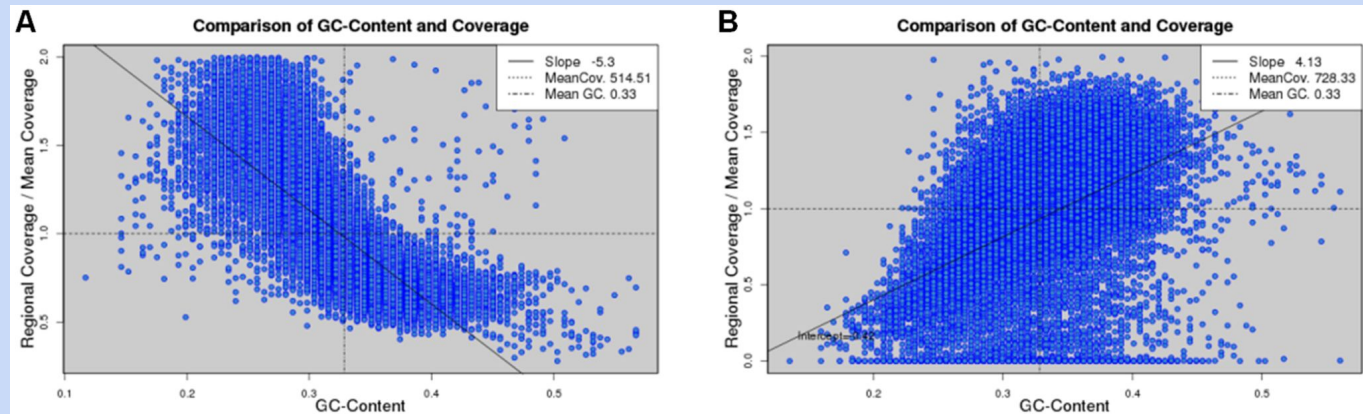
thermodynamics
and stronger
bonds
complicates
strand separation
for PCR

too much or
too little
amount of G or
C in a genomic
region

reduces PCR
amplification
efficiency and
sequencing
quality

**biased
coverage**

**artefacts that
cover or
mislead the
real biology**



Source (Chen 2013)

Where in the genome does the signal fall?

Ok, so you have a signal, but... where is it?

Is the distribution what we expect?

We are starting to move into the biological interpretation.

We expect:

- Both **H3K27ac** and **H3K4me1** should enrich regulatory regions, but not identically

- The **shared input** to be distributed evenly (so larger areas, such as intergenic regions, should get more of the total signal)

Categories that can be checked hierarchically:

- TSS (transcription start site)

- exons

- introns

- distal regulatory regions

- intergenic

Peak calling

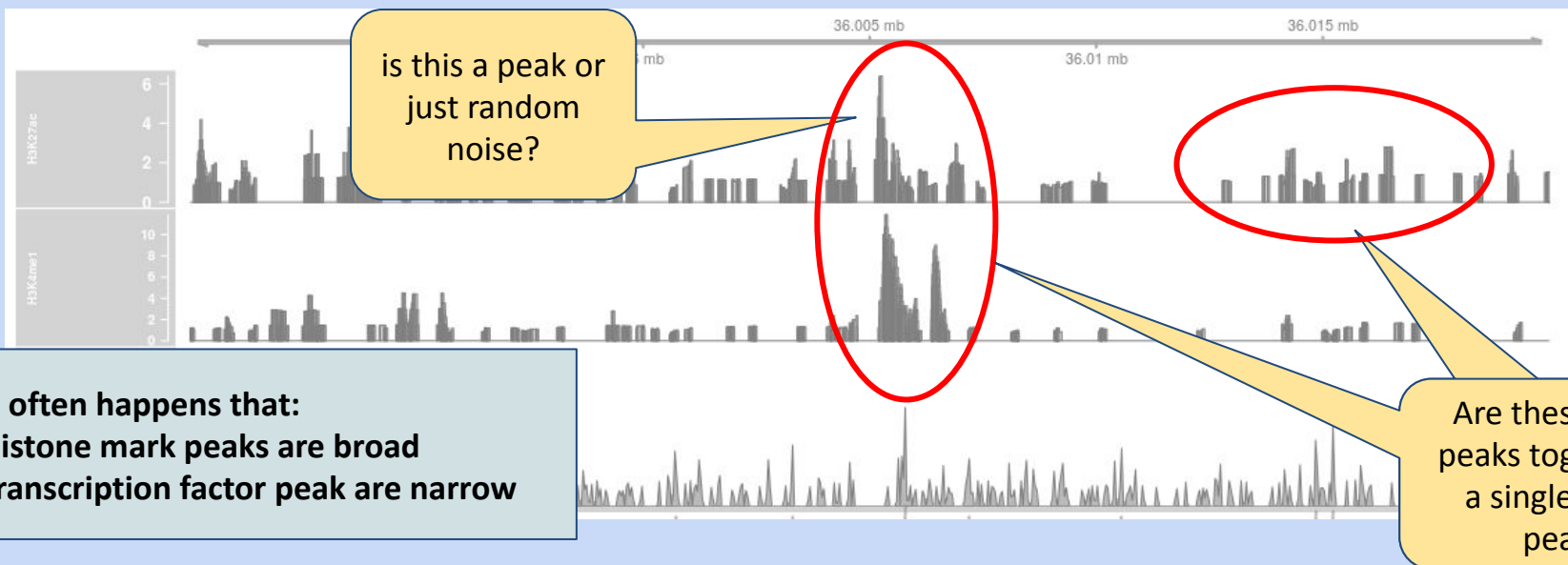
From aligned reads to enriched regions

Now we have beautiful signal that we start to believe has something... but there is still a lot of noise, so the question is:

What is a biologically meaningful peak, and what is just noise?

Is this part a single broad peak, or several short peaks?

These are mathematically challenging questions because **signal is sparse and peak width is unknown**. In today's practical we will use MACS2 to solve this.



Peak quality and biological integration

Not all peak sets are equally convincing!

We use **FRiP (Fraction of Reads in Peaks)** to measure how many of the reads are actually contributing to a peak.

We want this number to be as high as possible to reduce noise and capture signal.

In our practical we will calculate this directly in bash using *samtools* and *bedtools*:

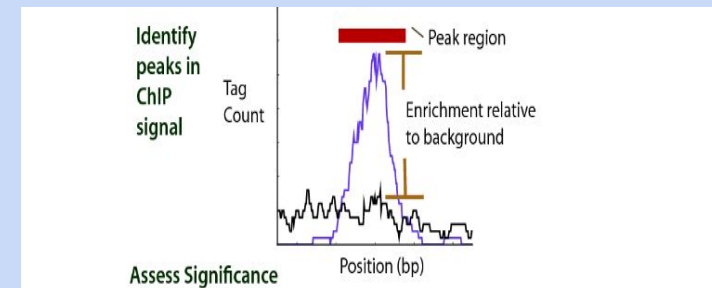
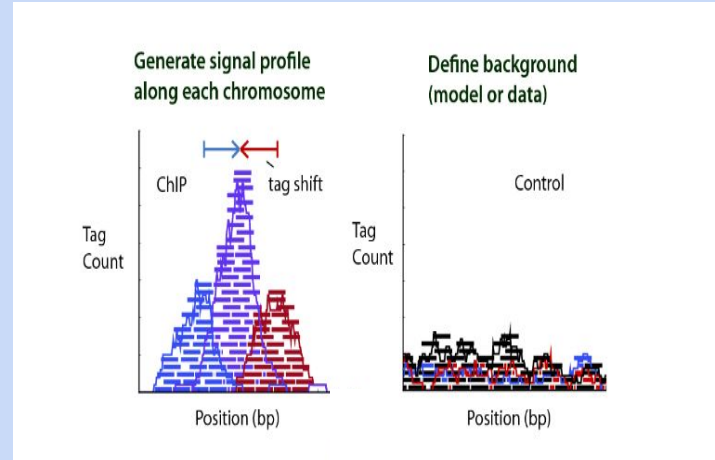
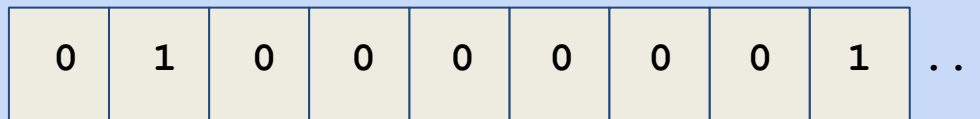
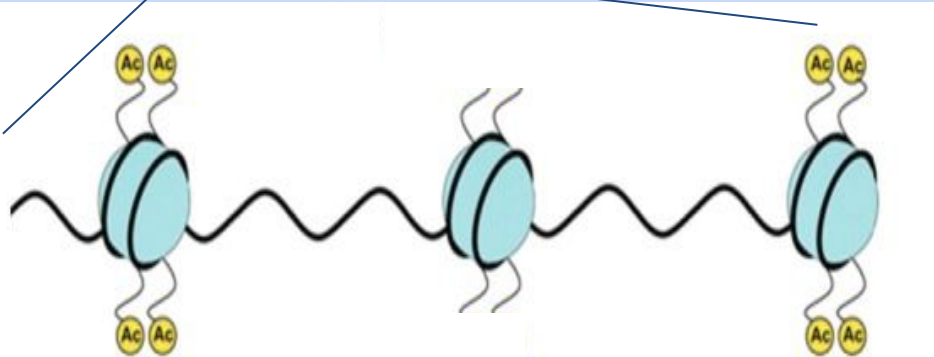
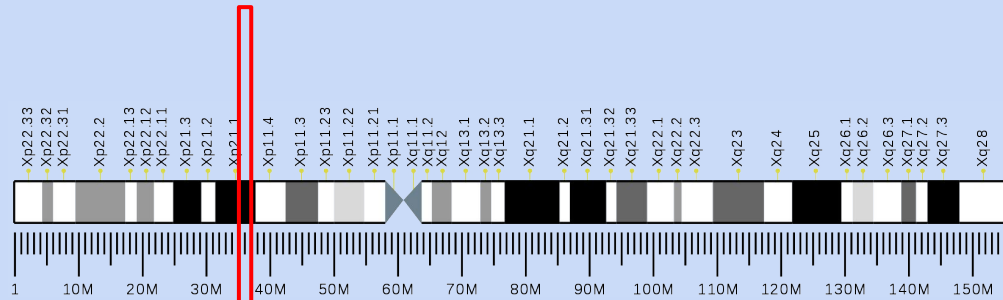
- Total reads > `samtools view -c -F 4`

- Reads in peaks > `bedtools coverage -counts`

- FRiP-like % > computed from those counts with `awk`

From local peaks to genome segmentation

ChromHMM and final interpretation



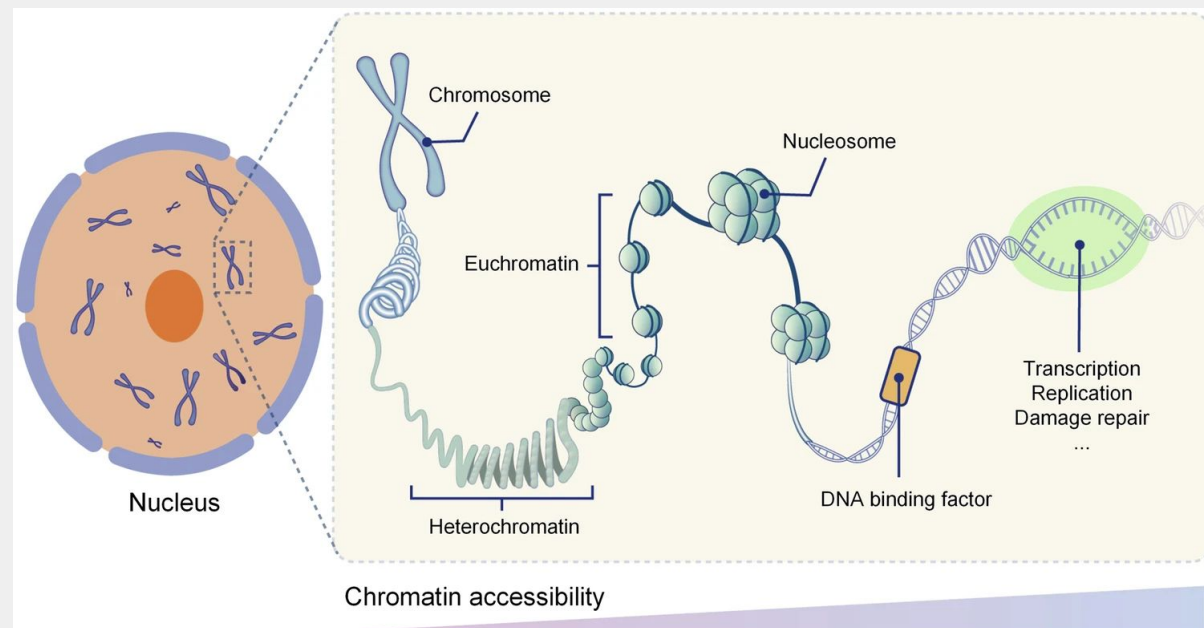
(Pepke 2009)

Using **ChromHMM** (Ernst 2012)



Take-home messages

1. Epigenomics explains regulatory differences beyond DNA sequence.
2. ChIP-seq measures genomic enrichment of a protein or chromatin mark.
3. Histone marks are interpreted combinatorially, not one by one.
4. ChIP-seq analysis needs ChIP-specific QC.
5. Peaks become more informative when integrated with chromatin states.



Let's play with it!

I propose ten tasks (with fancy names!)

If time is short, they can skip some of the red ones.

1. **Metaforge**: build the sample sheet
2. **Tilerhythm**: cluster samples in 1-kb windows
3. **Trackorama**: inspect browser tracks
4. **Strandscape**: compute strand cross-correlation
5. **GCBloom**: visualise GC bias
6. **Genomaplex**: summarise genomic signal distribution
7. **Peaksmith**: call broad peaks and assess peak quality
8. **Enhensemble**: intersect peaks and annotate candidate active regulatory regions
9. **Stateforge**: learn and interpret chromatin states with ChromHMM
10. **Karyoscape**: visualise candidate regulatory states along chr21